Lesson 1: Rectilinear motion of particles

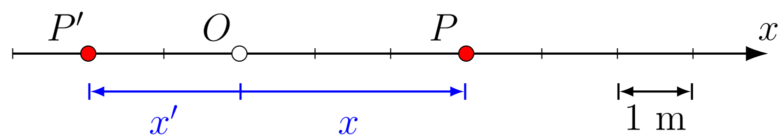
A particle that moves along a straight line is in **rectilinear motion**. The line can have any orientation in space (left-right, up-down, or some arbitrary angle). To fully describe the motion of a particle along a straight line, we would like to know three things about it: its position, velocity, and acceleration.

The goal of this lesson is to completely describe the motion of the particle as a function of time. For example, we might be provided with the position as a function of time and asked to find the particle's velocity and acceleration. Or we might be given the acceleration and asked to find the velocity and position of the particle.

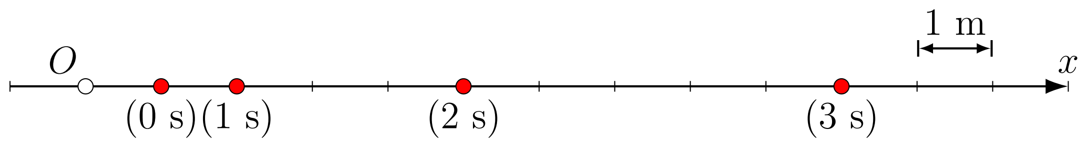
1 Position, Velocity, Acceleration

Position, Velocity, Acceleration

Consider a particle that is moving along the line shown below. We have picked a point for our origin (labeled O) and we have chosen some units of distance: each tick mark is 1 m. A particle at point P has a location of $x = 3$ m. The particle located at P' is at $x = -2$ m.



If we are given the position of the particle at any instant in time, we can say that the motion of a particle is known. Typically, this is achieved by having a mathematical expression for the position as a function of time. For example, we might be told that the position of a particle is given by $x(t) = 1 + t^2$. The line below shows the position of the particle at times of $t = 0, 1, 2, 3$ seconds.



Looking at the figure we can see that the particle is moving away from the origin. The average velocity of the particle is defined as the displacement in a given time interval divided by the duration of that time interval.

$$\text{Average velocity} = \frac{\Delta x}{\Delta t}$$

The table below computes some average velocities using the above example.

Time Interval (s)	Δx (m)	Δt (s)	Avg. Vel. (m/s)
0 – 1	1	1	1
0 – 2	4	2	2
0 – 3	9	3	3
1 – 2	3	1	3
1 – 3	8	2	4
2 – 3	5	1	5

Table 1: Calculation of average velocity for various time intervals for a particle with position given as $x(t) = 1 + t^2$.

The instantaneous velocity is determined by taking the time interval to be infinitesimally small.

$$\text{Instantaneous velocity} = v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t}$$

The table below computes the velocity of the above particle at $t = 2$ using smaller and smaller time intervals.

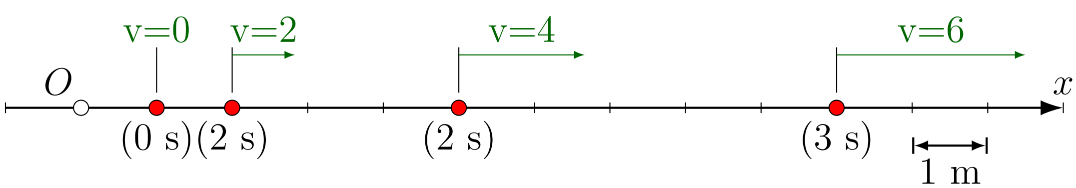
Time Interval (s)	Δx (m)	Δt (s)	Avg. Vel. (m/s)
2 – 2.20	0.840	0.2	4.20
2 – 2.10	0.410	0.1	4.10
2 – 2.01	0.0401	0.01	4.01
2 – 2.001	0.004001	0.001	4.001

Table 2: Calculation of velocity at $t = 2$ using smaller time intervals for a particle with position given as $x(t) = 1 + t^2$. The velocity is $v(t) = 2t$ which is 4 m/s at $t = 2$ s.

This infinitesimal limit is the definition of the derivative of position with respect to time. The velocity of a particle along a line is therefore given as

Velocity of a particle along a line

$$v = \frac{dx}{dt}$$



The figure below now includes the instantaneous velocity at a few given times. The average acceleration of the particle is defined as the change in velocity at a given time interval divided by the duration of that time interval.

$$\text{Average acceleration} = \frac{\Delta v}{\Delta t}$$

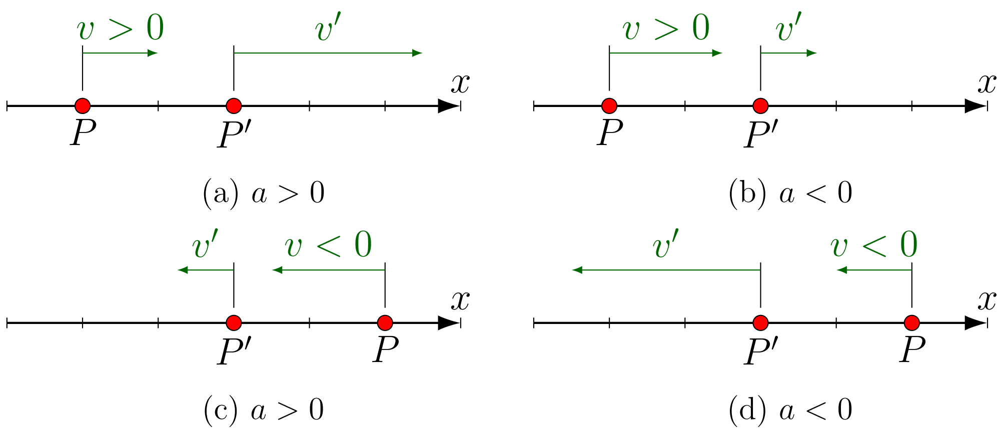
It turns out that for he particular example we have been looking at the average acceleration is equal to 2 m/s^2 regardless of which time interval we use. This is because the acceleration is a constant over the entire motion.

$$\text{Instantaneous acceleration} = a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t}$$

Acceleration of a particle along a line

$$a = \frac{dv}{dt}$$

1.1 A note on signs



(a) $a > 0$ (b) $a < 0$

(c) $a > 0$ (d) $a < 0$

Figure 1: A particle is initially at P with the velocity as shown. A small time intervale later, the particle is at location P' with a new velocity. For each of the four cases, determine if the acceleration is positive or negative.

Sample Problem 1.1

A particle moves along a straight line. Its position as a function of time is given as

$$x = t^3 - 3t^2 - 24t + 36$$

where x is in meters and t is in seconds. Find (a) the time when the

1.1 – cont

velocity is zero, (b) the position and the total distance traveled at that time, and (c) the acceleration at that time.

We first find the equations for the velocity and acceleration by taking derivatives

$$x = t^3 - 3t^2 - 24t + 36$$
$$v = \frac{dx}{dt} = 3t^2 - 6t - 24$$
$$a = \frac{dv}{dt} = 6t - 6$$

Part (a) asks us to find the time when $v = 0$.

$$0 = 3t^2 - 6t - 24 \longrightarrow t = -2 \text{ s and } t = 4 \text{ s}$$

Only the solution $t = 4$ s occurs after the motion has started.
The position at that time is

$$x(t = 4) = 4^3 - 3(4)^2 - 24(4) + 36 = -24 \text{ m}$$

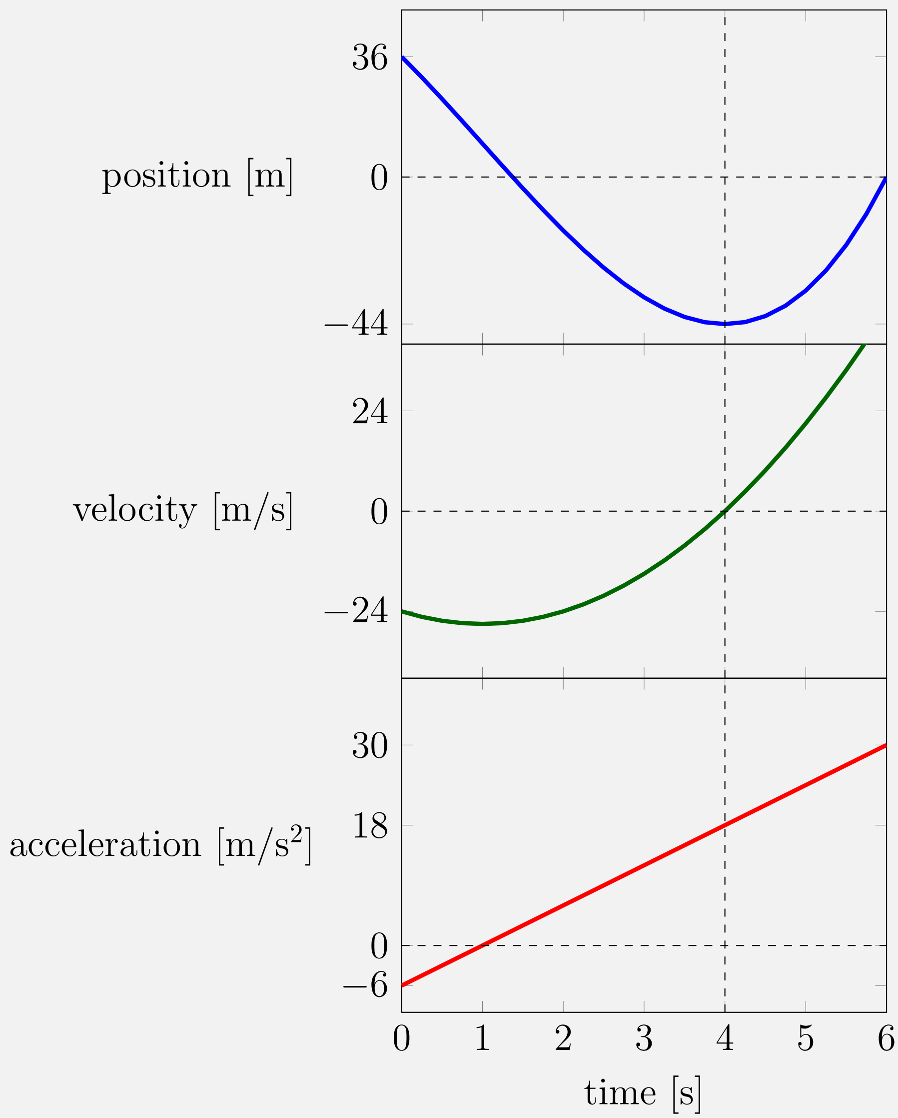
The distance traveled is the starting position minus the final position

$$\text{distance traveled} = x(4) - x(0) = -24 - 36 = -60 \text{ m}$$

The acceleration at that time is

$$a(t = 4) = 6(4) - 6 = 18 \text{ m/s}^2$$

1.1 – cont



position [m]
36
0
−44
velocity [m/s]
24
0
−24
acceleration [m/s²]
30
18
0
−6
0
1
2
3
4
5
6
time [s]

2 Uniform rectilinear motion

For this type of straight-line motion $a(t) = 0$ for all values of time. The velocity is therefore a constant for all time:

$$\frac{dx}{dt} = v = \text{constant}$$

To find the position, we integrate the above expression

$$\begin{aligned}\frac{dx}{dt} &= v \\ dx &= v\,dt \\ \int_{x_0}^{x} dx &= v\int_{0}^{t} dt \\ \boxed{x(t) = x_0 + vt}\end{aligned} \tag{1}$$

3 Uniformly accelerated rectilinear Motion

For this type of straight-line motion $a(t) = a = \text{constant}$ for all time values. We integrate to find the velocity. In what follows I've chosen to take $t_0 = 0$.

$$\begin{aligned}
\frac{dv}{dt} &= a = \text{constant} \\
dv &= a\,dt \\
\int_{v_0}^{v} dv &= a \int_0^t dt \\
v - v_0 &= at \\
\boxed{v(t) = v_0 + at}
\end{aligned} \tag{2}$$

We now integrate again

$$\begin{aligned}
\frac{dx}{dt} &= v_0 + at \\
dx &= (v_0 + at)\,dt \\
\int_{x_0}^{x} dx &= \int_0^t (v_0 + at)\,dt \\
x - x_0 &= v_0 t + \frac{1}{2}at^2 \\
\boxed{x(t) = x_0 + v_0 t + \frac{1}{2}at^2}
\end{aligned} \tag{3}$$

$$\begin{aligned}
v\frac{dv}{dx} &= a = \text{const} \\
v\,dv &= a\,dx \\
\int_{v_0}^{v} v\,dv &= a \int_{x_0}^{x} dx \\
\frac{v^2}{2} - \frac{v_0^2}{2} &= a\,(x - x_0) \\
\boxed{v^2(t) = v_0^2 + 2a\,(x - x_0)}
\end{aligned} \tag{4}$$

Sample Problem 1.1

A sprinter runs a 100-m race. For the first 5 seconds the runner accelerates uniformly and covers a distance of 25 meters. The runner then maintains a constant velocity for the remaining 75 meters. Determine the runner's a) acceleration, b) final velocity, c) finishing time.

For the first part of the race the runner accelerates uniformly and we can use eq. 3

$$x = x_0 + v_0 t + \frac{1}{2}at^2$$
$$25 = 0 + 0t + \frac{1}{2}a(5)^2 \Longrightarrow a = 2 \text{ m/s}^2$$

Now that we have the acceleration the velocity at the end of the first 5 seconds is found using eq. 2

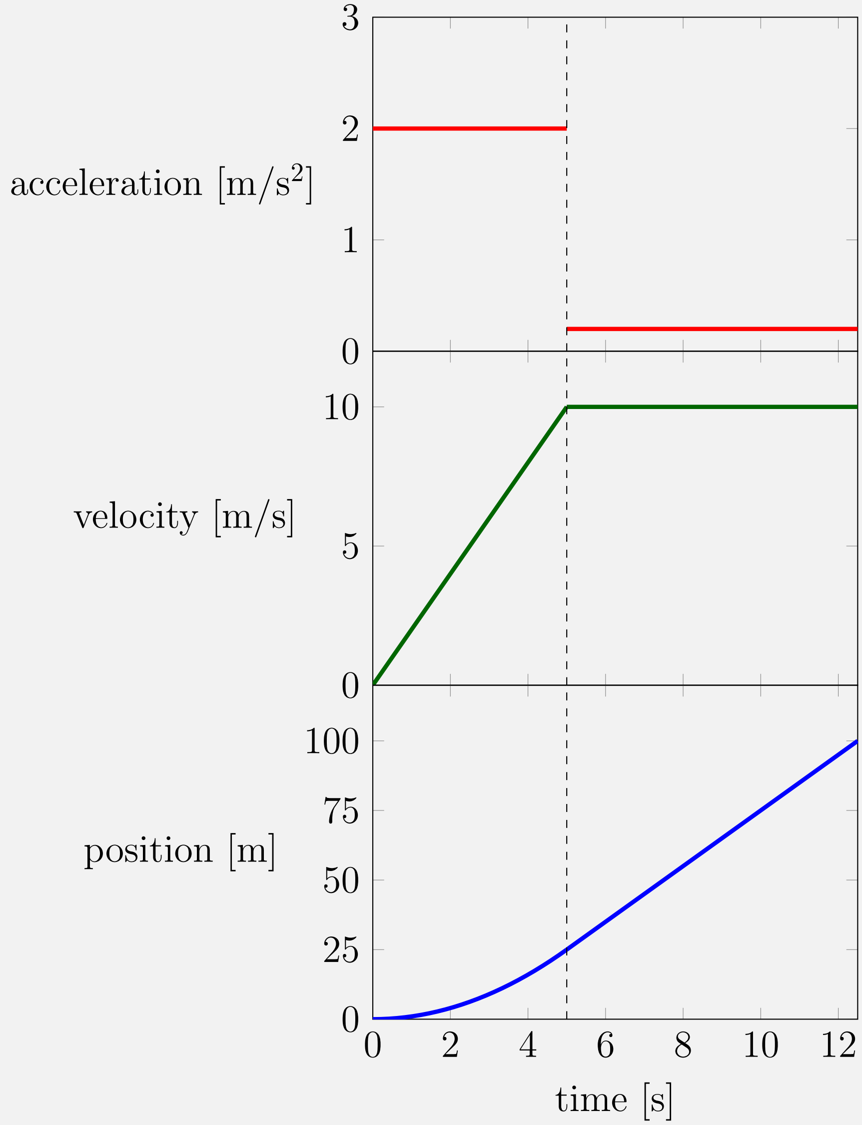
$$v = v_0 + at$$
$$v = 0 + 2(5) \Longrightarrow v = 10 \text{ m/s}$$

Since the velocity is unchanged after this point $v = 10\text{m/s}$ is also the final velocity. The time required to run the remaining 75 meters is found from eq. 1

$$x(t) = x_0 + vt$$
$$100 = 25 + 10t \Longrightarrow t = 7.5 \text{ s}$$

The total time is therefore the sum of the initial 5 second acceleration phase and 7.5 seconds of constant velocity for a total of 12.5 seconds.

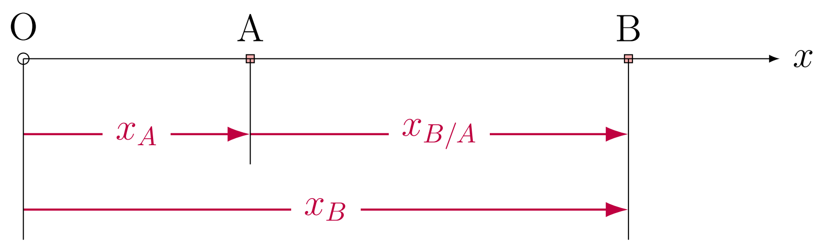
1.1 – cont



acceleration [m/s²]
velocity [m/s]
position [m]
time [s]
3
2
1
0
10
5
0
100
75
50
25
0
0 2 4 6 8 10 12

4 Relative and dependent motion of particles

Define $x_{B/A} = x_B - x_A$



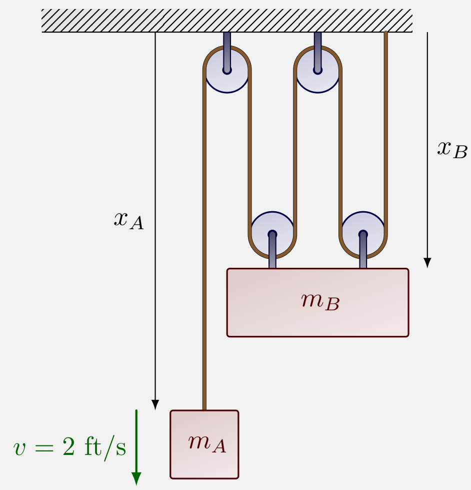
Sample Problem 1.1

In an elevator shaft, a ball is thrown vertically upward with an initial velocity of 18 m/s from a height of 12 m above the ground. At the same instant, an open-platform elevator passes the 5 m level, moving upward with a constant velocity of 2 m/s. Determine (a) when and where the ball hits the elevator, (b) the relative velocity of the ball with respect to the elevator when the ball hits the elevator.

Sample Problem 1.1

In the pulley system shown below mass A has a velocity of 2 ft/s at some instant in time. What is the upward speed of mass B at the same instant?

1.1 – cont



Let us call the total length of the rope L.

$$L = x_A + 4x_B$$

Taking the time derivative (noting the L is a constant)

$$0 = v_A + 4v_B$$
$$v_B = -\frac{1}{4}v_A$$
$$v_B = -\frac{1}{4}(2 \text{ ft/s})$$
$$v_B = -0.5 \text{ ft/s}$$

So, the velocity of mass B is 0.5 ft/s upward.

Sample Problems

1. A particle moves along a straight line. Its position as a function of time is given as

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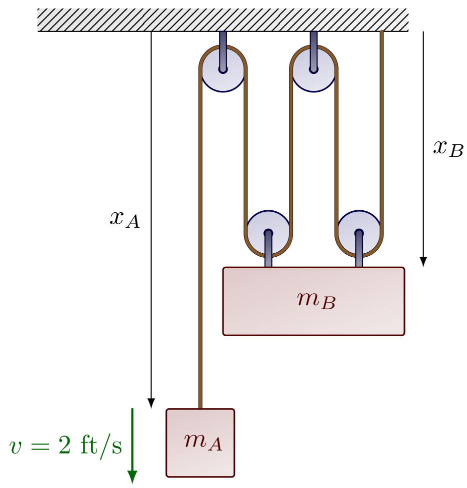
where x is in meters and t is in seconds. Find (a) the time when the velocity is zero, (b) the position and the total distance traveled at that time, and (c) the acceleration at that time.

2. A sprinter runs a 100-m race. For the first 5 seconds the runner accelerates uniformly and covers a distance of 25 meters. The runner then maintains a constant velocity for the remaining 75 meters. Determine the runner's a) acceleration, b) final velocity, c) finishing time.

3. In an elevator shaft, a ball is thrown vertically upward with an initial velocity of 18 m/s from a height of 12 m above the ground. At the same instant, an open-platform elevator passes the 5 m level, moving upward with a constant velocity of 2 m/s. Determine (a) when and where the ball hits the elevator, (b) the relative velocity of the ball with respect to the elevator when the ball hits the elevator.

4. In the pulley system shown below mass A has a velocity of 2 ft/s at some instant in time. What is the upward speed of mass B at the same instant?


x_A
x_B
m_B
m_A
$v = 2$ ft/s